The electromagnetic spectrum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electromagnetic wavelength, frequency and propagation.

SUCCESS CRITERIA - I can...

1. Use c=fλ in vacuum.
2. Order spectral regions by wavelength.
3. Distinguish electromagnetic from mechanical waves.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A closed-open pipe has length **0.3 m**.
1. Use λ=**4 L** for the fundamental and v=**340 m/s**.
2. **Find its fundamental frequency**.

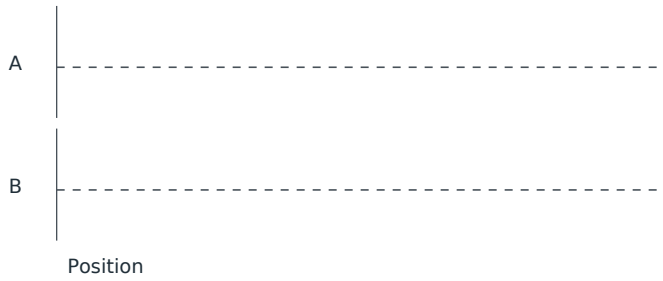
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: c = f λ; c = **3.0×10⁸ m/s**.
Electromagnetic waves in vacuum. Frequency identifies spectral region; mechanical sound needs a medium.

Worked reasoning: In vacuum, all electromagnetic waves travel at c=**3×10⁸ m/s**. A frequency of **10⁸ Hz** gives wavelength **3 m**. Higher frequency means shorter wavelength at the same speed.

Your turn: Draw two waves with a common speed but different wavelengths; **label which has higher frequency**.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Order radio, visible and X-rays from longest to shortest wavelength **0**. Which has greatest frequency in vacuum?

Radio	IR	Visible	UV	X-ray

Label increasing frequency and wavelength.

Marzano check - circle: ① ② ③

04 Apply the physics

Compare two signals in vacuum: light of wavelength **600 nm** and radio of frequency **100 MHz**. Use c = **3 × 10⁸ m/s**, **1 nm = 10⁻⁹ m** and **1 MHz = 10⁶ Hz**. **Find the light's frequency** and radio wavelength. **Complete the table**.

Signal	Frequency (Hz)	Wavelength (m)
Light	____	600×10^{-9}
Radio	100×10^{6}	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why can sunlight cross space but ordinary sound from a speaker cannot cross a vacuum?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A heat engine operates between **600 K** and **400 K** and absorbs **800 J**. **Find maximum ideal work** and **explain why actual work** is usually lower.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

07 Exit check

A wave has f=**2.4×10^{9} Hz** in vacuum.
1. Find λ.
2. **Name the spectral region**.

Radio	IR	Visible	UV	X-ray

Label increasing frequency and wavelength.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Visible light has wavelength **600 nm**.
1. **Find frequency** using **1 nm=10^{-9} m**.
2. Compare with light of wavelength **300 nm**.



Marzano check - circle: ① ② ③